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MIL-STD-2045-14502-1A
27 July 1995

MILITARY STANDARD

INFORMATION TECHNOLOGY DOD STANDARDIZED PROFILE INTERNET TRANSPORT PROFILE FOR DOD COMMUNICATIONS

Part 1: Transport and Internet Services

AMSC N/A

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Foreword

This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD). All previous versions of this document are hereby superseded.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this MIL-STD should be addressed to the:

Joint Interoperability and Engineering Organization (JIEO)
ATTN: JEBBF
Squier Hall, Bldg.. 283
Fort Monmouth, New Jersey 07703-5613

by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this MIL-STD or by memorandum.

This Defense Standardized Profile (DSP) is a functional DOD Data Communications Protocol Standard (DCPS) produced by the DCPS Technical Management Panel (DTMP). The MIL-STD-2045 document series was established within the DCPS Standardization Area to allow for the enhancement of commercial standards or the development of standards that are unique to DOD.

The MIL-STD-2045-10000 series, MIL-STD-2045-10000 to MIL-STD-2045-19999 inclusive, is used to describe how DOD implements commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Profiles for Open Systems Internetworking Technologies (POSIT) will serve as the base for developing the 10000 series with DOD enhancements. POSIT has superseded the Government Open Systems Interconnection Profile (GOSIP). Unique military standards will be used only when necessary.

The MIL-STD-2045-20000 series, MIL-STD-2045-20000 to MIL-STD-2045-29999 inclusive, is used to describe DOD enhancements and extensions to existing commercial, international, national, or federal standards.

The MIL-STD-2045-30000 series, MIL-STD-2045-30000 to MIL-STD-2045-39999 inclusive, is used to describe protocols and services unique to DOD that will not be supported by commercial, international, national, or federal standards.

The MIL-STD-2045-40000 series, MIL-STD-2045-40000 to MIL-STD-2045-49999 inclusive, is used to document interim standards. Interim standards document protocols and services needed by DOD until these protocols and services are described in either GOSIP (or POSIT) or in a MIL-STD-2045-20000 or -30000 series standard.

The MIL-STD-2045-50000 series, MIL-STD-2045-50000 to MIL-STD-2045-59999 inclusive, is used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required interim network services. The POSIT will serve as the base for developing the 50000 series, with DOD enhancements. Unique military standards will be used only when necessary.

Specific details and instructions for establishing a MIL-STD-2045 document, as well as profile development guidelines, are documented in MIL-HDBK-829A. DTMP Working Groups shall be responsible for DSP development and informal service and agency coordination; the DTMP Plenary shall be responsible for final review and approval.

This document is part of a set of DOD data communications protocol profiles based on the Internet protocol suite, and is intended to support the interoperability of DOD communication networks, including connectivity with the Defense Data Network (DDN).

The options and parameters specified in this DSP take precedence over MIL-STD 1777, Internet Protocol and MIL-

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STD 1778, Transmission Control Protocol. There are also tables in this DSP that contain the special Network and Transport layer requirements to fulfill the MIL-STD-188-220A Combat Net Radio (CNR) mission for interoperability of C³I tactical systems.

This part of MIL-STD 2045-14502 contains one normative annex and one informative annex:

Annex A (normative) DSPICS REQUIREMENTS LIST (DPRL).
Annex B (informative) CONCLUDING MATERIAL.

The Preparing Activity for this standard is the Data Communications Protocol Standards Technical Management Panel (DTMP). The custodians for this document are identified in the Defense Standardization Program, "Standardization Directory (SD-1)" and are classified in the Federal Supply Classification (FSC) system under Data Communications Protocol Standards (DCPS). Additional information can be obtained from:

Joint Interoperability and Engineering Organization
ATTN: DTMP Chairman
Squier Hall, Bldg. 283
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Introduction

This DOD Standardized Profile (DSP) is defined within the context of functional standardization, in accordance with the principles specified by MIL-HDBK-829A. The context of functional standardization is one part of the overall field of Information Technology (IT) standardization activities covering base standards, profiles, and registration mechanisms. A profile defines a combination of base standards that collectively perform a specific well-defined IT function. Profiles standardize the selection of options and other variations in the base standards to promote system interoperability and to provide a basis for the development of uniform, internationally recognized system tests.

One of the most important roles for a DSP is to serve as the basis for the development of recognized tests. DSPs also guide implementors in developing systems that fit the needs of the US Department Of Defense (DOD). DSPs are produced not simply to 'legitimize' a particular choice of base standards and options, but to promote real system interoperability. The development and widespread acceptance of tests based on this and other DSPs is crucial to the successful realization of this goal.

This MIL-STD is part of a set of Command and Control (C^2) common data communication profiles. It will be phased out upon the transition of the various military service and agency (S/A) networks to the Profiles for Systems Internetworking Technologies (POSIT). The purpose is to record what presently exists in, and what is needed, to achieve the interoperability of the various S/A data communication networks deployed to support a Joint Task Force (JTF).

This profile is part one of a six part transport profile. This part specifies the corresponding transport and network layers for the Transmission Control Protocol/Internet Protocol (TCP/IP) protocol suite. The transport layer consists of TCP and the User Datagram Protocol (UDP), and the network layer consists of Internet Protocol (IP) and Internet Control Message Protocol (ICMP). Additional requirements for the network and transport layers to support MIL-STD-188-220A, which specifies a protocol stack for interoperability of C^3I systems over Combat Net Radio (CNR), are also contained. The backward compatible enhancements to network and transport layer base standards in this DSP required for the operation of systems "in" or "across" CNR subnetworks are described in the appropriate PICS tables. These enhancements are optional for non-CNR systems.

This part of MIL-STD 2045-14502 contains one normative annex and one informative annex:

Annex A (normative) DSPICS REQUIREMENTS LIST (DPRL).
Annex B (informative) CONCLUDING MATERIAL.

The base standards specified in this DSP are Request For Comments (RFCs) designated as Official Internet Architecture Board (IAB) standards.

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Information Technology - DOD Standardized Profile (DSP) - Internet Transport Profile - Part 1: Transport and Internet Services

1 Scope

1.1 General

This DOD standardized profile applies to end systems concerned with implementing the TCP/IP protocol suite. It specifies a combination of protocols that provide a connectionless or a connection-oriented transport service over a connectionless network service.

1.2 Position within the taxonomy.

This profile contains requirements for both a connection-oriented and a connectionless transport service. It also contains requirements for a connectionless network service.

1.3 Scenario

This profile specifies the provision for connectionless or connection-oriented transport service between an end system connected to a subnetwork and another compatible end system through the IP connectionless network service. The interoperable end system may use mutually agreed upon access methods contained within this DSP, or may conform to a mutually agreed upon alternative access method. An end system is compatible only if the suboptions (e.g., TCP) are compatible. This profile contains the subnetwork independent requirements. A complete transport profile requires knowledge of the subnetwork type, access method, circuit type, and service type. The layer standards that make up this profile are shown in Figure 1.

ISO/TCP/IP Layers	Base Standard	Taxonomy
TRANSPORT LAYER	IAB STD 7 (TCP) IAB STD 6 (UDP) MIL-STD 188-220A. SELECTIVE DIRECTED BROADCAST - SEGMENTATION / REASSEMBLY	DSP TCP/IP Group Profile MIL-STD-2045-14502-1A
NETWORK LAYER	IP/ICMP IAB STD 5 (IP)	

Figure 1. Transport Profile Scenario

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2. References

2.1 Government Documents

2.1.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

MIL-HDBK 829A:	1 Sept 1994	*Guidelines for DOD Standardized Profiles.*
MIL-STD-2045-14502-2:	July 1994	*Point-to-Point Links*
MIL-STD-2045-14502-3:	July 1994	*Wide Area Network Access*
MIL-STD-2045-14502-4:	July 1994	*Local Area Network (LAN) Media Independent Requirements*
MIL-STD-2045-14502-5:	July 1994	*Carrier-Sense Multiple Access With Collision Detection (CSMA/CD) Local Area Network (LAN) Media Independent Requirements*
MIL-STD-2045-14502-6A:	27 July 1995	*Combat Net Radio (CNR)*

DOD activities may obtain copies of DOD directives through their own publication channels or from the DOD Single Stock Point, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other federal agencies and the public may purchase copies from the U.S. Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.

Copies of Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Others must request copies of FIPS from the National Technical Information Services, 5285 Port Royal, Springfield, VA 22161-2171.

2.1.2 Other Government documents, drawings and publications.

MIL-STD 188-220A:	27 July 1995	*Interoperability Standard for Digital Message Transfer Device Subsystems*

2.2 Non-Government publications

The following documents form a part of this document to the extent specified herein. Unless otherwise specified the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation.

2.2.1 Profiles

NONE

2.2.2 Base Standards

IAB STD 3	*(RFC 1122: 1989, Requirements For Internet Hosts - Communication Layers, RFC 1123: 1989, Requirements For Internet Hosts - Application and Support)*

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IAB STD 5	*(RFC 791: 1981, Internet Protocol, RFC 950: 1985, Internet standard subnetting procedure, RFC 919: 1984, Broadcasting Internet datagrams, RFC 922: 1984, Broadcasting Internet datagrams in the presence of subnets, RFC 792: 1981, Internet Control Message Protocol, RFC 1112: 1989, Host extensions for IP multicasting)*
IAB STD 6	*(RFC 768: 1980, User Datagram Protocol)*
IAB STD 7	*(RFC 793: 1981, Transmission Control Protocol)*
RFC 1108	*U.S. Department of Defense, Security Options for the Internet Protocol*

To obtain RFCs by mail write to the:

Network Information Center
14200 PARK Meadow Dr.
Suite 200
Chantilly, VA 22021

The NIC can be reached, by phone, Monday through Friday, 7 AM through 7 PM, Eastern Standard time:

1-800-365-3642
1-703-802-4535

RFCs may also be obtained from the DS.INTERNIC.NET via FTP, WAIS, and electronic mail. Through FTP, RFCs are stored as rfc/rfcnnnn.txt or rfc/rfcnnnn.ps where 'nnnn' is the RFC number, txt is a text file and ps is a postscript file. Login as "anonymous" and provide your e-mail address as the password.

Through WAIS, you may use either your local WAIS client or TELNET to DS.INTERNIC.NET and login as " wais" (no password required).

RFCs can also be obtained through electronic mail from the automated NIC mail server by sending a message tomailserv@ds.internic.net and include the following commands in the message body document -by-name RFC#### where #### is the RFC number without leading zeros or file /ftp/rfc/rfc####.yyy where 'yyy' is 'ps or 'txt.' To obtain the complete RFC index, the subject line of your message should read "RFC index."

2.2.3 Other Non Government documents, drawings, and publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation

"Round Trip Time Estimation," P. Karn & C. Partridge, ACM SIGCOMM-87, August 1987.

"Congestion Avoidance and Control," V. Jacobson, ACM SIGCOMM-88, August 1988.

"Congestion Control in IP/TCP," J. Nagle, RFC-896, January 1984.

2.3 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 Definitions

"Shall" means mandatory
"Shall not" means prohibited
"Piggyback" means attached to, or integrated with, another 'piece' or segment of information.

Internet Architecture Board (IAB) Standards (STD): The IAB has established this as an official standard protocol for the Internet. These protocols are assigned STD numbers and are based on one or more RFCs.

Request For Comments (RFC): RFCs are the working notes of the "Network Working Group," which is the Internet research and development community.

Note: All standards are published RFCs, but not all RFCs specify standards.

4 Abbreviations and Acronyms

ACK	ACKnowledgment
ALP	Application Layer Protocol
CNR	Combat Net Radio
DISA	Defense Information Systems Agency
DOD	Department of Defense
DPRL	DSPICS Requirements List
DSP	DOD Standardized Profile
DSPICS	DOD Standardized Profile Protocol Implementation Conformance Statement
GOSIP	Government Open Systems Interconnection Profile
IAB	Internet Architecture Board
ICMP	Internet Control Message Protocol
IHL	Internet Header Length
IP	Internet Protocol
ISN	Initial Sequence Number
ISO/IEC	International Organization for Standardization/International Electrotechnical Commission
IT	Information Technology
IUT	Implementation Under Test
JTF	Joint Task Force
MSS	Maximum Segment Size
OSI	Open Systems Interconnection
PICS	Protocol Implementation Conformance Statement
RFC	Request For Comments
S/A	Service/Agency
STD	Standard
SYN	Synchronization
TCP	Transmission Control Protocol
TOS	Type Of Service
TTL	Time To Live
UDP	User Datagram Protocol

5 Requirements

A conforming implementation of this profile shall be unconditionally compliant (as defined in RFC 1122, paragraph 1.3.2), and therefore, shall satisfy all the MUST and all the SHOULD requirements of the reference base standards. A conforming implementation of this profile shall not implement any capability that has been identified by the base standard as "SHOULD NOT" be implemented.

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5.1 General Requirements

Implementations claiming conformance to this DSP shall support the following elements as stated.

(a) All requirements in the remainder of section 5.
(b) All mandatory requirements of the standards referenced by this profile.

All of the constraints specified in Annex A (normative). In the case of Combat Net Radio, implementations shall conform to the modifications set forth in the DSPICS. Both general and specific modifications pertinent to Combat Net Radio are indicated in the DSPICS.

5.2 Transport Layer Requirements

A conforming implementation of this profile shall use TCP (IAB STD 7, RFC 793) and/or UDP (IAB STD 6, RFC 768), and shall conform to Requirements for Internet Hosts - Communication Layers (IAB STD 3, RFC 1122)

Paragraph 5 5 1.2 below describes an optional Segmentation/Reassembly Protocol that may be used over UDP to prevent fragmentation by the IP implementation in intermediate systems and routers. The Segmentation/Reassembly Protocol also provides for segment acknowledgment to support efficient, reliable delivery of large messages in a connectionless transport environment.

5.2.1 Transmission Control Protocol (TCP)

5.2.1.1 Major Capabilities

A conforming implementation of TCP shall support the following capabilities in accordance with the indicated base standards:

(a) data transfer as specified in RFC 793, sections 1.5, 2.8, and 3.7.
(b) addressing (especially well-known ports) as specified in RFC 793, section 2.7 and RFC 1122, section 4.2.2.1
(c) fragmentation and reassembly as specified in RFC 793, sections 1 5, 2 7, and 3 3 and RFC 1122, section 4.2.2.6.
(d) piggyback acknowledgment as specified in RFC 793, sections 1.5, 2.6, 3 7, and 3.9 and RFC 1122, section 4.2.3.2
(e) flow control as specified in RFC 793, section 1.5, 2.6, and 3.7 and RFC 1122, section 4.2.3.3.
(f) checksum computation as specified in RFC 793, sections 1.5, 3.1, and 3 8, and RFC 1122, section 4.2.2.7.
(g) error recovery as specified in RFC 793, sections 1.5 and 3.7, and RFC 1122, sections 4.2.3.1 and 4.2.2.15.
(h) precedence and security as specified in RFC 793, sections 1.5, 2.9, and 3.6.
(i) multiplexing as specified in RFC 793, sections 1.5, 2.7, and 3.8 and RFC 1122, section 4.2.2.1 and 4.2.2.18.

5.2.1.2 Header Format

A conforming implementation shall support the following fields as described in RFC 793, section 3.1:

(a) source port
(b) destination port
(c) sequence number

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(d) acknowledgment number
(e) data offset
(f) reserved field
(g) control bits (urgent, acknowledgment, push, reset, synchronize, finish data send)
(h) window field
(i) checksum
(j) urgent pointer
(k) options (end of option list, no-operation, maximum segment size)
(l) padding
(m) data

5.2.1.3 Detailed Requirements

This paragraph identifies additional requirements, or clarifications of requirements to support military use of Internet standards referenced by this DSP.

5.2.1.3.1 Interface Requirements

5.2.1.3.1.1 ICMP messages from IP

A conforming implementation shall act upon ICMP messages as stated in RFC 1122, section 4.2.3.9.

5.2.1.3.1.2 TCP/Application Layer Protocol (ALP) Interface

A conforming implementation shall include the features described in RFC 1122, section 4.2.4.

5.2.1.3.1.3 IP Options

A conforming implementation shall support the IP options delineated in RFC 1122, section 4.2.3.8.

5.2.1.3.1.4 Address Validation

A conforming implementation shall perform the functions described in RFC 1122, section 4.2.3.10.

5.2.1.3.2 Connection Management and Data Transfer Requirements

5.2.1.3.2.1 Push Flag

The push flag shall be implemented as described in RFC 793, section 2.8 and RFC 1122, section 4.2.2.2.

5.2.1.3.2.2 Window Management

Window management shall be implemented as described in RFC 793, sections 3.1 and 3.7, and RFC 1122, sections 4.2.2.3, 4.2.2.16, and 4.2.2.17

5.2.1.3.2.3 Urgent Data

Urgent Data shall be labeled as described in RFC 793, section 3.1, and RFC 1122, section 4.2.2.4.

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5.2.1.3.2.4 TCP Options

TCP Options shall be implemented as described in RFC 793, section 3.1, and RFC 1122, sections 4.2.2.5 , 4.2.2.6

5.2.1.3.2.5 TCP Checksums

A conforming implementation shall compute and check checksums as stated in RFC 793, section 3.1, and RFC 1122, section 4.2.2.7.

5.2.1.3.2.6 Initial Sequence Number (ISN) Selection

ISN selection shall be performed as described in RFC 793, section 3.3, and RFC 1122, section 4.2.2.9.

5.2.1.3.2.7 Opening Connections

The opening connection shall be implemented as described in RFC 793, section 3.4 and RFC 1122, sections 4 2.2.10, 4.2.2.11, 4.2.2.18, 4.2.3.7, and 4.2.3.10.

5.2.1.3.2.8 Closing Connection

Closing a connection shall be implemented as described in RFC 793, section 3.5 and RFC 1122, section 4.2 2.12 and 4.2.2.13.

5.2.1.3.2.9 Retransmissions

Retransmission shall be implemented as described in RFC 793 section 3.7 and RFC 1122, sections 4.2 2.15 and 4.2.3.1.

5.2.1.3.2.10 Generating ACKnowledgments (ACKs)

A conforming implementation shall generate ACKs as described in RFC 793, section 3.9, and RFC 1122, sections 4.2.2.20, 4.2.2.21, 4.2.3.2, and 4.2.3.3.

5.2.1.3.2.11 Sending Data

A conforming implementation shall perform the functions described in RFC 1122, sections 4.2.2.19, and 4.2.3 4 The Nagle algorithm shall be used, and implementations shall retain the capability to disable the Nagle algorithm when necessary.

5.2.1.3.2.12 Connection Failures

A conforming implementation shall act upon a connection failure as described in RFC 1122, section 4.2.3.5.

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5.2.1.3.2.13 Send Keep-alive Packets

Keep-alive packets are recommended for use in military implementations, but are not mandatory. There are no additional requirements to those described in RFC 1122, section 4.2.3.6.

5.2.2 User Datagram Protocol (UDP)

5.2.2.1 Major Capabilities

A *conforming implementation of UDP shall support the following:*

a) data transfer as specified in RFC 768, page 2 and RFC 1122, section 4.1.1.
b) port addressing as specified in RFC 768, pages 1 and 2 and RFC 1122, sections 4.1.1, 4.1.3.1, and 4.1.3.5, and 4.1.3.6.
c) checksum as specified in RFC 768, page 2 and RFC 1122, sections 4.1.1 and 4.1.3.4.

5.2.2.1.1 Header Format

A conforming implementation shall support the following fields as described in RFC 768:

(a) source port
(b) destination port
(c) length
(d) checksum

5.2.2.2 Detailed Requirements

5.2.2.2.1 Interface Requirements

General Interface requirements are as specified in RFC 768.

5.2.2.2.1.1 ICMP Messages from IP

A conforming implementation shall act upon all ICMP messages as stated in RFC 1122, section 4.1.3.3.

5.2.2.2.1.2 UDP/ALP Interface

A conforming implementation shall provide UDP Application Layer Services as described in RFC 1122, section 4.1.4.

5.2.2.2.1.3 IP Options

A conforming implementation shall pass IP options as described in RFC 1122, section 4.1.3.2. A user-defined IP option is described for use with multiple addressing. *This DSP defines and profiles this option.*

5.2.2.2.2 Data Transfer Requirements

5.2.2.2.2.1 UDP Checksum

A conforming implementation shall allow for a checksum as described in RFC 1122, section 4.1.3.4.

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5.2.2.2.2 UDP Multihoming

A conforming implementation shall provide for UDP Multihoming as described in RFC 1122, section 4.1 3 5.

5.3 Network Layer Requirements

A conforming implementation of this profile shall use IAB STD 5, specifically the Internet Protocol (IP) and the Internet Control Message Protocol (ICMP). A conforming implementation of the IP protocol shall conform to IAB STD 3, RFC 1122 (Requirements For Internet Hosts - Communication Layers).

The DOD Internet Protocol, IP, forms the network layer of this profile. IP is specified in RFC 791, as amended by RFC 950 (IP Subnet Extensions), RFC 919 (IP Broadcast Datagrams), and RFC 922 (IP Broadcast Datagrams with subnets). It also includes ICMP, the Internet Control Message Protocol, specified by RFC 792, which provides a mechanism for communicating control and error information between hosts or between hosts and gateways. Although ICMP is an integral part of IP, it uses the support of IP as if it (ICMP) were a higher layer protocol.

5.3.1 Internet Protocol

5.3.1.1 Major Capabilities

A conforming implementation shall support the following capabilities:

(a) data transfer as specified in RFC 791, sections 1.1 and 2.3.
(b) addressing as specified in RFC 791, sections 1.4, 2.3, 3.1, and 3.2 and RFC 1122, section 3.2.1.3.
(c) fragmentation/reassembly as specified in RFC 791, section 1.4, 2.3, 3.1, 3.2 and RFC 1122, section 3.3.2, 3.3.3, and 3.2.1.4.
(d) header checksum as specified in RFC 791, section 1.4, 3.1 (page 14), and RFC 1122, section 3.2.1.2.
(e) Type of Service (TOS) field as specified in RFC 791, section 1.4, 3.1, 3.2 and RFC 1122, section 3.2.1.6.
(f) time-to-live as described in RFC 1122, section 3.2.1.7
(g) additional options as described in RFC 1122, section 3.2.1.8, as further modified by this standard.

5.3.1.2 Datagram Format

A conforming implementation shall support the following fields as described in RFC 791, section 3.1 and 3.2:

(a) version number, also as specified in RFC 1122, section 3.2.1.1
(b) Internet Header Length (IHL)
(c) type of service (TOS), also as specified in RFC 1122, section 3.2.1.6
(d) total length of datagram
(e) segment identification, also as specified in RFC 1122, section 3.2.1.5
(f) control flags
(g) fragment offset, also as specified in RFC 791, section 3.2 and RFC 1122, sections 3.2.1.4 and 3.2.1.5
(h) time-to-live (TTL), also as specified in RFC 1122, section 3.2.1.7
(i) protocol
(j) header checksum, also as specified in RFC 1122, section 3.2.1.2
(k) source address, also as specified in RFC 1122, section 3.2.1.3
(l) destination address, also as specified in RFC 1122, section 3.2.1.3

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(m) IP options, also as specified in RFC 1122, section 3.2.1.8. Paragraph 5.5.3 of this DSP describes a user-defined option for use with multiple addresses.
(n) padding
(o) data

5.3.1.3 Detailed Requirements

5.3.1.3.1 Interface Requirements

5.3.1.3.1.1 Internet/Transport Layer Interface

The procedure calls described in RFC 791, section 3.3 and RFC 1122, section 3.1, 3.3.4, and 3.4 shall constitute the Internet/Transport Interface.

5.3.1.3.2 Data Transfer

5.3.1.3.2.1 Address Handling

A conforming implementation shall structure its addresses as a class A, B, C, or D address and shall follow the procedure described in RFC 1122, section 3.2.1.3

5.3.1.3.2.1.1 Routing Outbound Datagrams

A conforming implementation providing gateway functionality shall support the functions described in RFC 1122, section 3.3.1.

5.3.1.3.2.1.2 Local Multihoming

A conforming implementation shall provide for the requirements as described in RFC 1122, section 3.3.4.

5.3.1.3.2.1.3 Source Route Forwarding

A conforming implementation shall provide for the requirements as described in RFC 1122, section 3.3.5.

5.3.1.3.2.1.4 Broadcasts

A conforming implementation shall provide for the requirements as described in RFC 1122, section 3.3.6.

5.3.1.3.2.1.5 Multicasting

A conforming implementation shall provide for the requirements as described in RFC 1122, sections 3.2.3 and 3.3.7.

5.3.1.3.2.2 Fragmentation and Reassembly

A conforming implementation shall provide for the requirements as described in RFC 791, sections 3.1, 3.2, and RFC 1122, sections 3.2.1.4, 3.2.1.5, 3.3.2 and 3.3.3.

5.3.1.3.2.3 Type Of Service (TOS)

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A conforming implementation shall provide the TOS field as specified in RFC 791, sections 1.4, 3.1, 3.2, and RFC 1122, section 3.2.1.6.

5.3.1.3.2.4 Time-To-Live (TTL)

A conforming implementation shall provide the TTL field as specified in RFC 791, sections 3.1 , 3.2, and RFC 1122, section 3.2.1.7.

5.3.1.3.3 Options

All the options specified in RFC 791, sections 3.1, 3.2 and RFC 1122, section 3.2.1.8 shall be mandatory to implement but optional to use.

5.3.2 Internet Control Message Protocol (ICMP)

5.3.2.1 Major Capabilities

A conforming implementation shall support the following capabilities as described in RFC 792, "Introduction" (page 1), and RFC 1122, section 3.2.2.

(a) error reporting
(b) control messages

5.3.2.2 ICMP Message Format

ICMP messages are sent using the basic IP header. The first octet of the data portion of the datagram is an ICMP type field; the value of this field determines the format of the remaining data. Any field labeled "unused" is reserved for later extensions and shall be zero when sent, but receivers shall not use these fields (except to include them in the header checksum). Each ICMP message also contains a code field. ICMP uses the following IP header fields, as described in RFC 792, Message Format (pages 2-3):

(a) version
(b) Internet Header Length (IHL)
(c) type of service (TOS)
(d) total length
(e) identification
(f) flags
(g) fragment offset
(h) Time To Live (TTL)
(i) protocol
(j) checksum
(k) source address
(l) destination address

5.3.2.2.1 ICMP Messages

The following messages shall be implemented but are optional for use as described in RFC 792, pages 4-20 and RFC 1122, section 3.2.2:

(a) destination unreachable
(b) time exceeding
(c) parameter problem

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Check the source to verify that this is the current version before use.

(d) source quench
(e) redirect
(f) echo
(g) echo reply
(h) timestamp
(i) timestamp reply
(j) address mask request
(k) address mask reply
(l) information request messages
(m) information reply messages

5.3.2.3 Detailed Requirements

5.3.2.3.1 Error Reporting

A conforming implementation shall use the error messages described in RFC 792 and RFC 1122, sections 3.2.2.1 through 3.2.2.5.

5.3.2.3.1.1 ICMP Error Message Prohibition

A conforming host shall not send ICMP error messages as a result of receiving the messages described in RFC 1122, section 3.2.2.

5.3.2.3.2 Query Messaging

A conforming implementation shall use the Query Messages in accordance with the procedures described in RFC 792 and RFC 1122, section 3.2.2.6 through 3.2.2.9.

5.4 Internet Group Management Protocol (IGMP).

5.4.1 Major Capabilities.

Internet Group Management Protocol (IGMP), described in RFC 1112 and RFC 1122, is used by IP hosts to report their host membership to any immediately neighboring multicast routers. It lets all the systems on a network know which hosts currently belong to which multicast groups. IGMP is an asymmetric protocol and is specified from the point of view of a host. IGMP is an integral part of IP, and although optional in the general sense, is to be implemented by all hosts conforming to level 2 of the IP multicasting specification.

5.4.1.1 IGMP Message Format.

IGMP messages are transmitted in IP datagrams. The IGMP message is encapsulated within the IP datagram and is a fixed size of 8 bytes. The format for IGMP messages is contained in RFC 1112.

5.4.1.2 Version.

The Version field shall indicate which version of IGMP is being used. The current value is one (1).

5.4.1.3 Type.

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There are two types of IGMP messages of concern to hosts:

1 = Host Membership Query (Query sent by Multicast Router)

2 = Host Membership Report (Response sent by Host)

5.4.1.4 Unused Field.

This field is not used. It shall be set to zero when sent and ignored when received.

5.4.1.5 Checksum.

The checksum is the 16-bit one's complement of the one's complement sum of the 8-octet IGMP message. For computing the checksum, the checksum field is zeroed. (The checksum is calculated in the same manner as the ICMP checksum.)

5.4.1.6 Group Address.

The group address is a class "D" IP address. In a query the group address is set to zero, and in a report it contains the 32-bit group address being reported.

5.5 Requirements for Combat Net Radio

The following paragraphs describe modifications to Annex A of this DSP that are applicable to systems operating in or across Combat Net Radio (CNR) subnetworks. THESE MODIFICATIONS, WHILE NECESSARY FOR USE WITH CNR, ARE NOT LIMITED TO CNR, AND ARE RECOMMENDED FOR GENERAL USE WHERE NEEDED The following paragraphs describe deviations or enhancements to IP, TCP, or UDP. Only deviations or enhancements to this DSP are found in the following paragraphs. CNR physical, data link, and Intranet layers are profiled in MIL-STD-2045-14502-6A.

5.5.1 Transport Layer CNR additions

5.5.1.1 Message Segmentation/Reassembly

When required, the message segmentation and reassembly capability shall be supported within the end systems.

5.5.1.2 Message Segmentation/Reassembly Protocol.

Message Segmentation/reassembly is a protocol intended for use with certain military applications running over UDP. Messages larger than the maximum segment size (MSS) permitted for UDP shall be segmented and reassembled in accordance with the procedures set forth in the following paragraphs. Messages that are smaller than the MSS do not require segmentation/reassembly and do not carry the segmentation/reassembly header unless an end-to-end acknowledgment is required.

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5.5.1.3 Overall operation.

The messages which are larger than the designated MSS shall be segmented by the source device prior to transmission, and reassembled at the destination device prior to delivery to the application. An end-to-end acknowledgment shall be employed to ensure reliable delivery of all segments in a connectionless transport environment. Upon successful receipt of all segments, the message shall be reassembled and transferred to the application layer. *Figure 2 provides an example of the Segmentation/Reassembly header.*

5.5.1.3.1 Maximum Segment Size (MSS).

The MSS shall be calculated using the equation below with the following variables as defined.

MMTU	- minimum Message Transfer Unit size of the Internet
SH	- segmentation/reassembly header size
UDP	- UDP header size
IP	- IP header size
MSS =	MMTU - (SH + UDP + IP)

The maximum message transmission unit (MTU) size defined for a CNR network, and which is provided to IP, shall be the data link frame information field size minus the maximum intranet header size (132 octets).

The following components take on maximized constant values based on the definitions provided within this section.
SH = 8 octets
UDP = 8 octets
IP = 60 octets

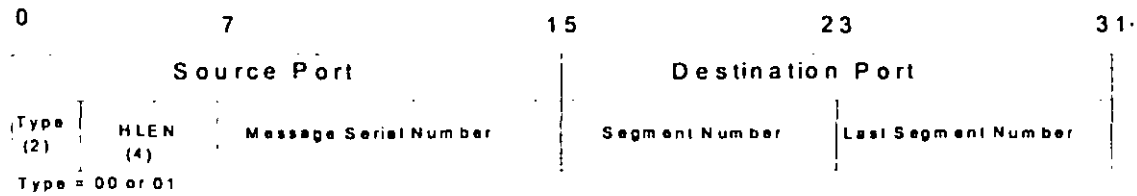


Figure 2. Segmentation / Reassembly Header

It is desirable that IP datagrams which will be transmitted across multiple subnetworks do not exceed 576 octets. An MSS of 500 octets will assure that IP fragmentation will not occur at IP router/gateway devices.

5.5.1.3.2 End-to-end acknowledgment.

A partial acknowledgment scheme shall be employed in which correctly received segments are acknowledged, even if all segments comprising a message have not been received. Every segmentation/reassembly header shall contain a last segment number from which the destination device may estimate the time required to receive all segments of a given message. A receive timer is set during which time it is expected that all segments of a message will be received correctly. When this timer expires, all segments correctly received shall be acknowledged through a bitmap in the segmentation/reassembly header. The source device shall only retransmit those segments which are not acknowledged.

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5.5.1.3.3 Segmentation/Reassembly header.

Three variants of segmentation/reassembly headers are provided which follow the structure presented in Figures 1, 2, and 3.

5.5.1.4 Source Port.

This port number identifies the originating upper layer process of which this message segment forms a part.

5.5.1.5 Destination Port.

This port number identifies the upper layer process for which this segment forms a part.

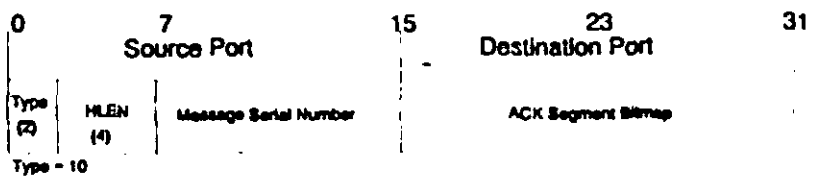
5.5.2 Type of Header.

This field identifies the type of header in use in accordance with the two-bit sequences below

Bits	Interpretation
00	End-to-end Acknowledgment required
01	End-to-end Acknowledgment not required
10	ACK segment bit map present
11	Complete ACK (bit map not required)

These bits indicate whether the header is a segmentation header or an acknowledgment header. In the case of a segmentation header, the second bit indicates whether this message requires end-to-end acknowledgments by all addressees. When an acknowledgment header is indicated the second bit indicates whether the acknowledgment is partial or complete. If the acknowledgment is partial, an ACK segment bit map follows the message serial number in which each message segment correctly received is indicated by setting a bit in the bitmap (e.g., first bit set and second bit at zero indicates the first segment was correctly received while the second segment was not correctly received). A complete acknowledgment means all segments of a message were received correctly, in which case an ACK segment bit map is not needed.

Figure 3. Partial Acknowledgment Header



5.5.2.1 Header Length (HLEN).

This field indicates the total length of the segmentation/reassembly header in 32-bit words. This value is used to determine the total length of the ACK segment bit map field in the acknowledgment header.

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5.5.2.2 Message Serial Number.

This number is assigned by the originating device and uniquely identifies the message of which this segment forms a part.

5.5.2.3 Segment Number.

This number identifies the segment's position in the whole message and is assigned by the originating device. It is used in the reassembly process.

5.5.2.4 Last Segment Number.

This number indicates the total number of segments comprising the message identified by the message serial number. It is used to set the message receipt timer and allow for partial acknowledgments when this timer expires prior to receipt of all message segments.

5.5.2.5 Acknowledged segments bitmap.

The bits in this field are used to indicate which segments of a message have been successfully received at the destination addressee. These bits are relative to the segment position in the whole message (e.g., bit no. 1 means segment no.1). A bit set (binary 1) means the segment has been correctly received. Binary 0 indicates the segment was not correctly received. Any segment numbers not relevant to the whole message are indicated as binary 0. This field is extensible in 32-bit increments, up to an additional thirteen 32-bit words. Unused bits added to pad the field to a 32-bit boundary shall be set to 1.

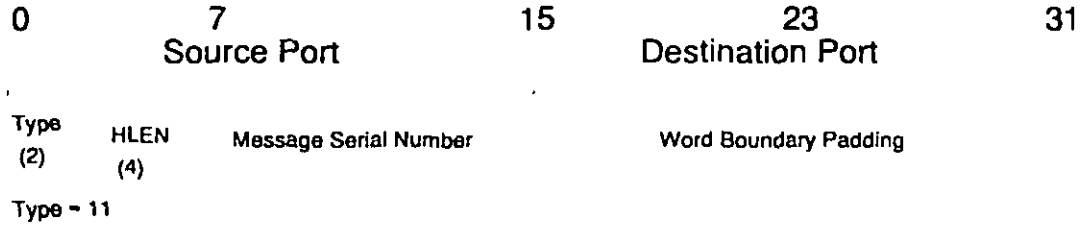


Figure 4 Acknowledged Segments Bitmap

5.5.2.6 Interface with other layers.

The message segmentation/reassembly protocol interfaces with the application layer protocol (ALP) and the protocol in the next layer down, which is the User Datagram Protocol (UDP) in the Transport Layer. The parameters exchanged between both the upper and lower layer protocols are the same as the parameters defined by the UDP upper layer protocol interface. Two of the UDP-specified interface parameters are the source and destination ports. If the message segmentation/reassembly header is present, the destination port number shall be a well-known, unique port number (assigned by the network administrator). The source port number provided to UDP shall be the same port number provided by the upper layer protocol to the segmentation/reassembly protocol. If the message segmentation/reassembly header is not present, the source

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and destination port numbers provided to UDP shall be the same as those provided by the upper layer protocol to the message segmentation/reassembly protocol.

5.5.2.7 Source device procedures (Segmentation).

All messages received from the application which exceed the MSS shall be segmented. The term "message" includes message body and message header. The message is segmented and a segmentation/reassembly header is appended to each segment. A message serial number shall be assigned and copied into the header of each segment of that message. Each segment is then sequentially numbered starting with 1. A retransmit timer is set if end-to-end acknowledgment is requested. If this timer expires without the receipt of a partial or complete acknowledgment for message segments, the entire message is retransmitted once. If the timer should expire without the receipt of at least a partial acknowledgment, the message shall be deleted and an error indication returned to the ULP or application. When a partial acknowledgment is received, only those segments which are not acknowledged shall be retransmitted. The message originator shall indicate in the segmentation header whether this message requires an end-to-end acknowledgment (Type field = 00) or does not require an end-to-end acknowledgment (Type field = 01).

5.5.2.8 Destination device procedures (Reassembly).

Upon receipt of a segment of a new message, the recipient shall set an accumulation timer during which it expects to receive a complete set of segments for the message. A destination device shall accumulate segments of a message with the same message serial number from the same source IP address and reassemble that message when all segments have been successfully received, using segment numbers to indicate segment position relative to the whole message. Segmentation/reassembly headers shall be removed. The whole message is then passed to the upper layer process indicated by the destination port number. If the accumulation timer expires before receiving all segments of a message, a partial acknowledgment (Type field 10) will be sent indicating 'where' in the segment ACK bitmap which segments of the message have been correctly received. If all segments of a message have been received prior to expiration of the accumulation timer's timeout period, the timer shall be stopped and a complete acknowledgment (Type field 11) shall be returned which indicates all segments of the message have been correctly received.

5.5.3 Selective Directed Broadcast - IP Multi-Address Feature

The efficient transmission of messages to multiple destinations is accomplished with the selective directed broadcast protocol or IP multicast (IPMC) feature. The selective directed broadcast protocol is implemented at the interface to the upper layers.

5.5.3.1 Applicable documents.

This protocol assumes the implementation of an Internet Protocol and User Datagram Protocol compliant with the other portions of this DSP (MIL-STD-2045-14502-1A).

5.5.3.2 Multi-Addressing Overview.

Transmissions to multiple destinations shall be accomplished with the selective directed broadcast protocol. The selective directed broadcast protocol is implemented at the interface between the upper layer process and either the segmentation/reassembly protocol (if used) or the User Datagram Protocol. The selective directed broadcast protocol uses UDP and the IP option field to pass multiple destination addresses. Therefore, the UDP protocol implementation must pass IP options as specified in RFC 1122, section 4.1.3.2. Unitdata primitives are used to pass information for the sending and receiving of data between the Segmentation/Reassembly layer and the Application Layer Protocol (ALP). For CNR, the selective directed broadcast protocol shall be used.

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5.5.3.3 Multi-address IP Option.

The Multi-address IP Option provides an ability for a host to list the intended destinations for a selective directed broadcast as shown in Figure 5. This option field is variable in length, up to a maximum of 38 octets due to the limitation of the Internet Header Length field.

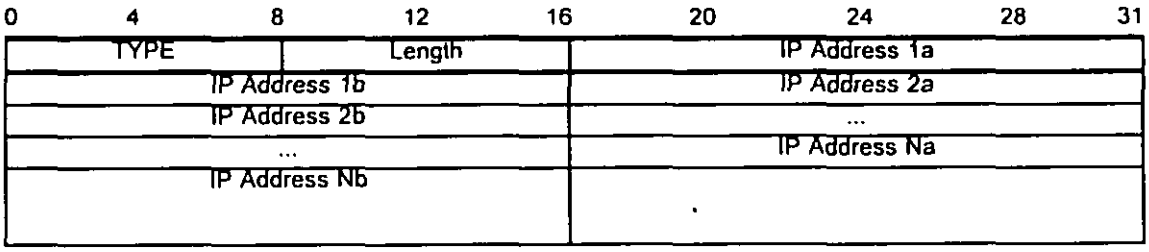


Figure 5. Layout of Multi-Address IP Option Field

a. The TYPE field specifies the copy flag, class and number of the option. The class and number are to be determined. The copy flag indicates whether or not this option field is to be copied into each fragment if an IP datagram is fragmented. The format of the type field is shown in Figure 6:

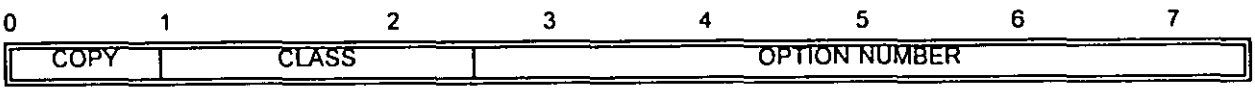


Figure 6. Type Field Layout

b. Since the multi-address option list shall always be copied to all IP headers during fragmentation COPY should be set to 1.

c. The class field shall be zero, indicating a control option.

d. The LENGTH octet indicates how many octets are in this option field. It is calculated as follows:

LENGTH = 2 + 4*(number IP addresses)

e. The remaining octets contain the IP addresses of destination hosts. Each IP address occupies 4 octets.

5.5.3.4 Sending Selective Directed Broadcast.

Selective directed broadcast is an Internet compliant method for an IP source host to deliver a single packet to several IP destination hosts using directed broadcast as specified in RFC 1122 and through use of an IP "Multi-address" option field. The multi-address IP option field will contain a list of up to nine IP addresses that are the intended recipients of the directed broadcast message.

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5.5.3.5 Procedure.

Procedures for source hosts, transmit routers and destination routers are provided below.

a. When a source host has a message to send to multiple destination hosts, it shall:

1. Group the destination host's Internet addresses by their network identifiers (Net IDs) If there are N distinct Net IDs there will be at least N distinct directed broadcast (DB) packets.

2 For each Net ID, form the directed broadcast address, as defined in RFC 1122, for that network. The DB address is used as the Destination Address in the IP packet and the Source Address is the address of the host sending the message.

3. Place the entire IP address for up to nine intended hosts in the group in the Multi-address IP option field. The total length of all IP options in a given datagram is limited to 40 octets as determined by the HLEN (Header Length) field in the IP header which measures the number of 32 bit words in the header. If other options are to be included in the IP header in addition to the multi-address option, the number of addresses in the multi-address option may need to be reduced.

4. Transmit the completed packets normally.

b. Transit routers shall route the packets normally. A destination gateway on a MIL-STD-188-220 network shall:

1. Form an Intranet protocol header containing the destination host's data link address and the relay addresses.
2. Transmit the packet within the network.

5.5.3.6 Receiving Selective Directed Broadcast.

In processing received selective directed broadcast datagrams, receiving hosts shall look inside the multi-address IP option field for their address. Processing shall continue only if a host's address is found inside this option field Thus, the source has explicit control in determining which hosts will process its datagrams. Since selective directed broadcast uses a broadcast address in the destination field, it can only be used with UDP, and not TCP at layer 4, because TCP is a point to point vice a point to multi-point protocol.

5.5.3.7 Intranet Protocol.

The Intranet layer, layer 3a, has been dedicated to routing intranet packets between a source and possibly multiple destinations within the same radio network. The intranet layer also accommodates the exchange of topology and connectivity information. The Intranet protocol profile description and protocol requirements are contained in Part 6 to this DSP, MIL-STD-2045-14502-6A.

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ANNEX A (normative)

DSPICS REQUIREMENTS LIST (DPRL)

A.1 Introduction

This document provides the DOD Standardized Profile Implementation Conformance Statements (DSPICS) Requirements List (DPRL) for implementations of MIL-STD-2045-14502-1A. The DSPICS for an implementation is generated by completing the DPRL in accordance with the following instructions.

An implementation shall satisfy the mandatory conformance requirements of the base standards referenced in this profile.

An implementation's completed DPRL is called the DSPICS. The DSPICS states which capabilities and options of the protocol have been implemented. The following can use the DSPICS:

(a) The protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight.

(b) The supplier and acquirer or potential acquirer of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard DSPICS proforma.

(c) The user or potential user of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to internetwork can often be predicted from incompatible DSPICSs).

(d) A protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.1.1 Notation

The following notations and symbols from MIL-HDBK 829A, which references ISO/IEC TR 10000-1 and -2, are used in the DPRL to indicate the status of features:

Status Symbols

m	- mandatory.
m <n>	- support of every item of the group labeled by the same numeral <n> required, but only one is active at a time.
o	- optional.
o.<n>	- optional, but support of at least one of the group of options labeled by the same numeral <n> is required.
c	- conditional.
-	- non-applicable (i.e. logically impossible in the scope of the profile).
x	- excluded or prohibited.
i	- out of scope of profile (left as an implementation choice).

In addition, the symbol "-" is used to indicate an option whose status is not constrained by the profile (status in the base standard). The o.<n> notation is used to show a set of selectable options (i.e., one or more of the set must be implemented) with the same identifier <n>

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Two character combinations may be used for dynamic conformance requirements. In this case, the first character refers to the static (implementation) status, and the second refers to the dynamic (use); thus "mo" means "mandatory to be implemented, optional to be used."

Notations for Conditional Status

The following predicate notations are used:

<predicate>:: This notation introduces a group of items, all of which are conditional on <predicate>.

<predicate>: This notation introduces a single item which is conditional on <predicate>.

In each case, the predicate may identify a profile feature, or a boolean combination of predicates. ("^" is the symbol for logical negation.)

<index>: This predicate symbol means that the status following it applies only when the DSPICS states that the features identified by the index are supported. In the simplest case, <index> is the identifying tag of a single DSPICS item. The symbol <index> also may be a Boolean expression composed of several indices.

<index>:: When this group predicate is true, the associated clause should be completed.

Notations used in the Protocol Feature Column

<r> *Symbol used to denote the receiving system.*
<t> Symbol used to denote the transmitting system.

If the protocol support "clause" column is blank, this means there is no change to the procedure or condition described in the referenced base standard. If a clause paragraph number is present, it may or may not indicate a significant change, and the reviewer is directed to refer to that clause in the DSP.

Support Column Symbols

The support of every item as claimed by the implementor is stated by circling the appropriate answer (Yes, No, or N/A) in the support column:

Yes	Supported by the implementation.
No	Not supported by the implementation.
N/A	Not applicable.

Base standard requirements are shown using the equivalent notations in upper case (e.g., M, O, X).

The classification of the requirements and options in the Internet RFCs do not correspond to the convention described in A.1.1 above, and have been mapped into the DSPICS as follows:

RFC	(MIL-STD) DSPICS
MUST	Mandatory
SHOULD	Mandatory
MAY	Optional
SHOULD NOT	Prohibited
MUST NOT	Prohibited

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A.1.2 Footnotes

Footnotes to the proforma are indicated by superscript numerals. The footnote appears on the page of the first occurrence of the numeral. Subsequent occurrences of a numeral refer to the footnote of the first occurrence.

A.1.3 Instructions for Completing the DPRL

A DSP implementor shows the extent of compliance to a DSP by completing the DPRL; that is, compliance to all mandatory requirements and the options that are not supported are shown. The resulting completed DPRL is called a DSPICS. Where this profile refines the features of the base standards, the requirements expressed in this DPRL shall be applied (as indicated in DPRL items with no "Profile Support" column) to constrain the allowable responses in the base standard DSPICS proforma. When this profile makes additional requirements, the "Profile Support" column for such DPRLs shall be completed. In this column, each response shall be selected either from the indicated set of responses, or it shall comprise one or more parameter values as requested. If a conditional requirement is inapplicable, use the Not Applicable (NA) choice. If a mandatory requirement is not satisfied, exception information must be supplied by entering a reference Xi, where i is a unique identifier, to an accompanying rationale for the noncompliance. When the profile requirement is expressed as a two-character combination (as defined in A.1.1 above), the response shall address each element of the requirement; e.g., for the requirement "mo," the possible compliant responses are "yy" or "yn."

A.2 Standards Referenced

This profile specifies the provision of TCP service in an End System from which a standardized IP network service operating over point-to-point circuits is available (or can be made available). It uses the following standards.

IAB STD 3	*(RFC 1122: 1989, Requirements For Internet Hosts - Communication Layers, RFC 1123: 1989, Requirements For Internet Hosts - Application and Support)*
IAB STD 5	*(RFC 791: 1981, Internet Protocol, RFC 950: 1985, Internet standard subnetting procedure, RFC 919: 1984, Broadcasting Internet datagrams, RFC 922: 1984, Broadcasting Internet datagrams in the presence of subnets, RFC 792: 1981, Internet Control Message Protocol, RFC 1112: 1989, Host extensions for IP multicasting)*
IAB STD 6	*(RFC 768: 1980, User Datagram Protocol)*
IAB STD 7	*(RFC 793: 1981, Transmission Control Protocol)*
RFC 1108	*Security Options for the Internet Protocol*

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A.3 DSPICS Requirements List (DPRL)

A.3.1 General Information

A.3.1.1 Identification of PICS

Ref	Question	Response
1	Date of Statement (DD/MM/YY)	
2	PICS serial number	
3	System Conformance statement cross-reference	

A.3.1.2 Identification of Implementation Under Test (IUT)

Ref	Question	Response
1	Implementation name	
2	Implementation version	
3	Machine name	
4	Machine version	
5	Operating System name	
6	Operating System version	
7	Special Configuration	
8	Other Information	

A.3.1.3 Global Statement of Conformance

Ref	Question	Response
1	Are all mandatory IAB STD requirements met for:	
2	IAB STD 3 (Internet Hosts)	
3	IAB STD 6 (UDP)	
4	IAB STD 7 (TCP)	
5	IAB STD 5 (IP,ICMP, IGMP) NOTE: IGMP is not required by this profile. However if IGMP is implemented, there are mandatory provisions called out in this profile.	

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A.4 Basic Requirements

The following table lists the detailed requirements for a TCP/IP implementation, and asks if the listed protocols have been implemented:

Index	Protocol	Clause of Profile	Profile Status	Support
tcp	IAB STD 7 and IAB STD 3, RFC 1122, section 4.2, TCP, implemented?	5.2	mo	Yes yes/no
udp	IAB STD 6 and IAB STD 3, RFC1122, section 4.1, UDP, implemented?	5.2	mo	Yes yes/no
ip	IAB STD 3 and IAB STD 3, RFC 1122, section 3, IP, implemented?	5.3	m	Yes
icmp	IAB STD 5 and IAB STD 3, RFC 1122, section 3, ICMP, implemented?	5.3	m	Yes
inhost	IAB STD 3, Communication Layers, implemented?	5.2	m	Yes

A.4.1 PICS Proforma for Transmission Control Protocol (TCP)

A.4.1.1 Identification

Supplier	
Contact Point for Queries	
Implementation name(s) and Versions	
Other Information Necessary for full identification - eg, name(s) and version(s) for machines and/or operating systems; System Name(s)	

A.4.1.2 Protocol Summary

Protocol Version	
Addenda Implemented?	
Amendments Implemented?	
Have any exceptions been required? (Note: A YES answer means that the implementation does not conform to the Transmission Control Protocol. Non-supported mandatory capabilities are to be identified in the PICS, with an explanation of why the implementation is non-conforming.	Yes_____ No_____
Date of Statement	

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A.4.1.3 General/Major Capabilities

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
dt	Data Transfer	RFC 793:1.5, 2.8,3.7	M	5.2.1.1	m	Yes
pa	Addressing	RFC 793: 2.7 RFC 1122: 4.2.2.1	M		m	Yes
frg	Fragmentation	RFC 793: 1.5, 2.2 RFC 1122: 4.2.2.6	M		m	Yes
ras	Reassembly		M		m	Yes
pa	Piggyback Acknowledgment	RFC 793: 1.5,2.8,3.7,3.9 RFC 1122: 4.2.3.2	M		m	Yes
flc	Flow Control	RFC 793: 1.5,2.8,3.7 RFC 1122: 4.2.3.3	M		m	Yes
chks	Checksum	RFC 793: 1.5,3.1 RFC 1122: 4.2.2.7	M		m	Yes
errec	Error Recovery	RFC 793: 1.5,3.7 RFC1122: 4.2.3.1, 4.2.2.15	M		m	Yes
secpr	Precedence and Security	RFC 793. 1.5, 2.9	M		m	Yes

A.4.1.4 Interfaces

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
icmp	Internet Control Message Protocol	RFC 1122. 4.2.3.9.	M	5.2.1.3.1	m	Yes
alp	TCP/Application Layer Protocol	RFC 1122: 4.2.4.	tcp:M		tcp:m	Yes N/A
ip	Internet Protocol Options	RFC 1122: 4.2.3.8	M		m	Yes
av	Remote Address Validation	RFC 1122: 4.2.3.10	M		m	Yes

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A.4.1.5 Frame Structure

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
hdr	Header Format	RFC 793: 3 1	M	5 2 1 2	m	Yes
sp	Source Port		M		m	Yes
dp	Destination Port		M		m	Yes
sn	Sequence Number		M		m	Yes
an	Acknowledgment Number		M		m	Yes
dof	Data Offset		M		m	Yes
res	Reserved field (must be zero)		M		m	Yes
Control Bits: (following six fields)						
urg	Urgent Pointer field - significant	RFC 793: 3 1 RFC1122 4 2 2 4	M	5 2 1 2	m	Yes
ack	Acknowledgment field - significant	RFC 793: 3 1	M		m	Yes
psh	Push function	RFC 793: 3 1 RFC1122 4 2 2 2	M		m	Yes
rst	Reset the connection	RFC 793: 3 1	M		m	Yes
syn	Synchronize sequence numbers		M		m	Yes
fin	No more data from sender		M		m	Yes
win	Window	RFC 793: 3 1 RFC1122 4 2 2 3 4 2 2 18, 4 2 2 17	M		m	Yes
cks	Checksum	RFC 793: 3 1 RFC 1122 4 2 2 7	M		m	Yes
opt	Options	RFC 793: 3 1 RFC1122;4 2 2 5, 4 2 2 6	M		m	Yes
eol	End of Option List	RFC 793: 3 1	M		mo	Yes yes/no
nop	No-Operation		M		mo	Yes yes/no
mss	Maximum Segment size		M		mo	Yes yes/no
pad	Padding		M		mo	Yes yes/no

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A.4.1.6 TCP Procedure

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
mux	Multiplexing	RFC 793: 1.5, 2.7, 3.8 RFC 1122: 4.2.2.1, 4.2.2.18	M	5.2.1.1	m	Yes
isnsel	Use clock-driven ISN selection	RFC 793: 3.3 RFC 1122: 4.2.2.9	M	5.2.1.3.2.6	m	Yes
	Opening Connections					
op1	Support simultaneous open attempts	RFC 793: 3.4 RFC 1122: 4.2.2.10	M	5.2.1.3.2.7	m	Yes
op2	SYN-RCVD remembers last state	RFC 793: 3.4 RFC 1122: 4.2.2.11	M		m	Yes
op3	Passive open call interfere with others	RFC 793: 3.8 RFC 1122: 4.2.2.18	x		x	No
op4	Function: simultaneous LISTENs for same port	RFC 793: 3.8 RFC 1122: 4.2.2.18	M		m	Yes
op5	Ask IP for source address for SYN if necc.	RFC 1122: 4.2.3.7	M		m	Yes
op5a	Otherwise, use local address of connection	RFC 1122: 4.2.3.7	M		m	Yes
op6	OPEN to broadcast/multicast IP Address	RFC 1122: 4.2.3.10	X		x	No
op7	Silently discard segment to broadcast/multicast address		M		m	Yes
	Closing Connections:					
cl1	RST can contain data	RFC 1122: 4.2.2.12	O	5.2.1.3.2.8	m	Yes
cl2	Inform application if connection is aborted	RFC 793: 3.4	M		m	Yes
cl3	Half-duplex close connections	RFC 1122: 4.2.2.13	O		o	Yes No
cl31	Send RST to indicate data lost		O		m	Yes
cl4	In TIME-WAIT state for 2xMSL seconds		M		m	Yes
cl41	Accept SYN from TIME-WAIT state		O		o	Yes No
	Retransmissions:					
rtc	Retransmission Timeout Calculation	RFC 793, 1.5,3.7, RFC 1122, 4.2.3.1 , 4.2.2.15	M	5.2.1.3.2.9	m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
rt1	Jacobsen slow start algorithm	RFC 1122 4.2.2.15	M	5.2.13.2.9	m	Yes
rt2	Jacobson congestion avoidance algorithm		M		m	Yes
rt3	Retransmit with same IP ident		O		o	Yes No
rt4	Karn's algorithm	RFC 1122 4.2.3.1	M		m	Yes
rt5	Jacobson's RTO estimation algorithm		M		m	Yes
rt6	Exponential Backoff		M		m	Yes
rt7	SYN RTO calculation same as data		O		m	Yes
rt8	Recommended initial values and bounds		O		m	Yes
	Generating ACK's:					
ga1	Queue out of order segment	RFC 1122 4.2.2.20	O	5.2.13.2.10	m	Yes
ga2	Process all queued before send ACK		M		m	Yes
ga3	Send ACK for out of order segment	RFC 1122 4.2.2.21	O		o	Yes No
ga4	Delayed ACK's	RFC 1122 4.2.3.2	O		m	Yes
ga41	Delay < 0.5 seconds		ga4:M		m	Yes
ga42	Every 2nd full sized segment ACK'd		ga4:M		m	Yes
ga5	Receiver SWS-Avoidance Algorithm	RFC 1122 4.2.3.3	M		m	Yes
	Sending Data:					
sd1	Configurable TTL	RFC 1122 4.2.2.19	M	5.2.13.2.11	m	Yes
sd2	Sender SWS-avoidance algorithm	RFC 1122 4.2.3.4	M		m	Yes
sd3	Nagle algorithm		O		m	Yes
sd31	Application can disable Nagle algorithm		sd3 M		m	Yes
	Connection Failures					
cf1	Negative advice to IP on R1 retransmissions	RFC 1122 4.2.3.5	M	5.2.13.2.12	m	Yes
cf2	Close connection on R2 retransmissions	RFC 1122 4.2.3.5	M		m	Yes
cf3	Application Layer protocol can set R2	RFC 1122 4.2.3.5	M		m	Yes
cf4	Inform Application Layer protocol of R1<=retransmissions<R2	RFC 1122 4.2.3.5	O		m	Yes
cf5	Recommend values for R1, R2	RFC 1122 4.2.3.5	O		m	Yes
cf6	Same mechanism for SYNs	RFC 1122 4.2.3.5	M		m	Yes
cf6a	R2 at least 3 minute for SYN	RFC 1122 4.2.3.5	M		m	Yes
cf6b	Send Keep alive packets	RFC 1122 4.2.3.6	O	5.2.13.2.13	o	Yes No
ka1	Application can request	RFC 1122 4.2.3.6	M		m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
ka2	Default is "off"	RFC 1122: 4.2.3.6	M	5.2.13.2.13	m	Yes
ka3	Only send if idle for interval		M		m	Yes
ka4	Interval configurable		M		m	Yes
ka5	Default at least 2 Hrs		M		m	Yes
ka6	Implementation must be tolerant of lost ACKs		M		m	Yes

A.4.2 PICS Proforma for User Datagram Protocol (UDP)

A.4.2.1 Identification

Supplier	
Contact Point for Queries	
Implementation name(s) and Versions	
Other Information Necessary for full identification - eg, name(s) and version(s) for machines and/or operating systems; System Name(s)	

A.4.2.2 Protocol Summary

Protocol Version	
Addenda Implemented	
Amendments Implemented	
Have any exceptions been required? (Note: A YES answer means that the implementation does not conform to the Transmission Control Protocol/User Datagram Protocol. Non-supported mandatory capabilities are to be identified in the PICS, with an explanation of why the implementation is non-conforming.	Yes _____ No _____
Date of Statement	

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A.4.2.3 General/Major Capabilities

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
dt	Data Transfer	RFC 768, RFC 1122 4 1 1	M	5.2.2.1	m	Yes
pa	Port Addressing	RFC 768, RFC 1122 4 1 1, 4 1 3 1, 4 1 3 5, 4 1 3 6	M		m	Yes
uchks	Checksum	RFC 768 RFC 1122, 4 1 1, 4 1 3 4	M		m	Yes

A.4.2.4 Basic Interfaces

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
uuc	User/UDP Interface	RFC 768	M	5 2 2 2 1	m	Yes
urp	Creation of new receive ports	RFC 768	M		m	Yes
usend	Send operations that specify data, source and destination ports	RFC 768	M		m	Yes
urecc	Receive operations that return the data octets and source port and source address	RFC 768	M		m	Yes
ulli1	Determine Source Internet Address	RFC 768	M		m	Yes
ulli2	Determine Destination Internet Address	RFC 768	M		m	Yes

A.4.2.5 Frame Structure

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
uhf	Header Format	RFC 768	M	5.2.2.1 1	m	Yes
usp	Source Port	RFC 768	M		m	Yes
udp	Destination Port	RFC 768	M		m	Yes
ucb	Length	RFC 768	M		m	Yes
ucks	Checksum	RFC 768	M		m	Yes

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A.4.2.6 UDP Procedure

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
us	UDP Send port unreachable	RFC 1122: 4.1.3.1	O	5.2.2.1	m	Yes
IP options in UDP:						
iu1	Pass received IP options to application layer	RFC 1122: 4.1.3.2	M	5.2.2.2.1.3	m	Yes
iu2	Application layer can specify IP options to Send	RFC 1122: 4.1.3.2	M		m	Yes
iu3	UDP passes IP options down to IP layer	RFC 1122: 4.1.3.2	M		m	Yes
icm	Pass ICMP msgs up to application layer	RFC 1122: 4.1.3.3	M		m	Yes
UDP checksums:						
uc1	Able to generate/check checksum	RFC 1122: 4.1.3.4	M	5.2.2.2.2.1	m	Yes
uc2	Silently discard bad checksum	RFC 1122: 4.1.3.4	M		m	Yes
uc3	Sender Option to not generate checksum	RFC 1122: 4.1.3.4	O		o	Yes No
uc31	Default is to checksum	RFC 1122: 4.1.3.4	M		m	Yes
uc4	Receiver Option to require checksum	RFC 1122: 4.1.3.4	O		o	Yes No
UDP Multihoming:						
um1	Pass specific-dest address to application	RFC 1122: 4.1.3.5	M	5.2.2.2.2.2	m	Yes
um2	Application Layer can specify local IP addr	RFC 1122: 4.1.3.5	M		m	Yes
um3	Application layer specify wild local IP addr	RFC 1122: 4.1.3.5	M		m	Yes
um4	Application layer notified of local IP address used	RFC 1122: 4.1.3.5	O		m	Yes
bip	Bad IP source address silently discarded by UDP/IP	RFC 1122: 4.1.3.6	M		m	Yes
vip	Only send valid IP source address	RFC 1122: 4.1.3.6	M		m	Yes
UDP Application Interface Services:						
ua1	Full IP interface described in section 3.4 of RFC 1122 for application	RFC 1122: 4.1.4	M	5.2.2.2.1.2	m	Yes
ua11	Able to specify TTL, TOS, IP options when sending	RFC 1122: 4.1.4	M		m	Yes
ua12	Pass received TOS up to application layer	RFC 1122: 4.1.4	O	5.2.2.2.1.2	o	Yes No

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A.4.3 PICS Proforma for Internet Protocol (IP)

A.4.3.1 Identification

Supplier	
Contact Point for Queries	
Implementation name(s) and Versions	
Other Information Necessary for full identification - eg, name(s) and version(s) for machines and/or operating systems; System Name(s)	

A.4.3.2 Protocol Summary

Protocol Version	
Addenda Implemented	
Amendments Implemented	
Have any exceptions been required? (Note: A YES answer means that the implementation does not conform to the Internet Protocol. Non-supported mandatory capabilities are to be identified in the PICS, with an explanation of why the implementation is non-conforming.	Yes_____ No_____
Date of Statement	

A.4.3.3 General/Major Capabilities

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
dt	Data Transfer	RFC 791: 1.1, 2.3	M	5.3.1.1	m	Yes
add	Addressing	RFC 791: 1.4, 2.3, 3.1, 3.2, RFC 1122: 3.2.1.3	M		m	Yes
ada	Class A addresses supported		M		m	Yes
adb	Class B addresses supported		M		m	Yes
adc	Class C addresses supported		M		m	Yes
frg	Fragmentation and Reassembly	RFC 791: 1.4, 3.1,2.3, 3.2, RFC 1122: 3.2.1.4	M	5.3.1.1	m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
frg1	Forward datagrams of 68 octets without fragmentation	RFC 791: 3.2 RFC 1122: 3.2.1.4	M		m	Yes
frg2	Receive datagrams of 576 octets either in one piece or in fragments		M		m	Yes
frg3	Don't Fragment Supported		M		m	Yes
reass1	IP Datagram reassembly supported	RFC 1122: 3.3.2	M		m	Yes
reass2	Reassembly timeout period supported		M		m	Yes
tos	Type Of Service	RFC 791: 1.4, 3.1, 3.2 RFC 1122: 3.2.1.6	M		m	Yes
tpre	Precedence	RFC 791: 3.1,3.2 RFC 1122: 3.2.1.6	M		m	Yes
tdel	Delay	RFC 791: 3.1,3.2	M		m	Yes
tthr	Throughput		M		m	Yes
trel	Reliability		M		m	Yes
ttl	Time To Live	RFC 791: 3.1,3.2 RFC 1122: 3.2.1.7	M		m	Yes
opt	Options	RFC 791: 3.1,3.2 RFC 3.2.1.8	M		m	Yes
chk	Checksum	RFC 791: 1.4, 3.1, 3.2 RFC 1122: 3.2.1.2	M		m	Yes

A.4.3.4 Interfaces

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
lni	Local Network Interface	RFC 791: 3.3, RFC 1122: 3.4	M	5.3.1.3.1.1	m	Yes
uli	Upper Level Interface	RFC 791: 3.3, RFC 1122: 3.4	M		m	Yes

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A.4.3.5 Frame Structure

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
Internet/Transport Layer Interface						
iii	Implement IP and ICMP	RFC 1122: 3.1	M	5.3.1.3.1.1	m	Yes
rm	Handle remote multihoming in application layer		M		m	Yes
lm	Support local multihoming		O		o	Yes No
lm1	Devices with multiple physical interfaces shall implement local multi-homing	RFC 1122 3.3.4	O		m	Yes
gs	Meet gateway specs if forward datagrams	RFC 1122 3.1	M		m	Yes
cs	Configuration switch for embedded gateway		M		m	Yes
cs1	Config. switch default to non-gateway		M		m	Yes
cs2	Automatic move by host into gateway mode		X		x	No
ldd	Able to log discarded datagrams		O		m	Yes
ldd1	Record in counter		O		m	Yes
sdv	Silently discard version not equal to 4	RFC 1122 3.2.1.1	M		m	Yes
vchk	Verify IP checksum, silently discard bad datagram	RFC 1122: 3.2.1.2	M		m	Yes
in1	Allow transport layer to use all IP mechanisms	RFC 1122 3.4	M		m	Yes
in2	Pass interface ident up to transport layer	RFC 1122 3.4	O		m	Yes
in3	Pass all IP options up to transport layer	RFC 1122 3.4	M		m	Yes
in4	Transport layer can send certain ICMP messages	RFC 1122 3.4	M		m	Yes
in5	Pass specified ICMP messages up to transport layer	RFC 1122 3.4	M		m	Yes
in51	Include IP header + 8 octets or more from originator	RFC 1122. 3.4	M		m	Yes
Addressing:						
add1	Subnet and Class A, B, and C special addressing restrictions (RFC 950)	RFC 1122 3.2.1.3	M	5.3.1.3.2.1	m	Yes
add2	Source address must be host's own IP address	RFC 1122: 3.2.1.3	M		m	Yes
add3	Silently discard datagram with bad destination address	RFC 1122: 3.2.1.3	M		m	Yes
add4	Silently discard datagram with bad source address	RFC 1122: 3.2.1.3	M		m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
Datagram Fragmentation and Reassembly:						
rea	Support datagram fragmentation and reassembly	RFC 1122: 3.2.1.4	M	5.3.1.3.2	m	Yes
idf	Retain same ID field in identical datagram	RFC 1122: 3.2.1.5	O		o	Yes No
TOS:						
tos1	Allow transport layer to set TOS	RFC 1122: 3.2.1.6	M	5.3.1.3.2.3	m	Yes
tos2	Pass received TOS up to transport layer	RFC 1122: 3.2.1.6	O		m	Yes
tos3	Use RFC 975 link layer mappings for TOS	RFC 1122: 3.2.1.6	O		x	No
TTL:						
ttl1	Send packet with TTL of 0	RFC 1122: 3.2.1.7	X	5.3.1.3.2.4	x	No
ttl2	Discard received packets with TTL < 2	RFC 1122: 3.2.1.7	X		x	No
ttl3	Allow transport layer to set TTL	RFC 1122: 3.2.1.7	M		m	Yes
ttl4	Fixed TTL is configurable	RFC 1122: 3.2.1.7	M		m	Yes
IP Options:						
op1b	Allow transport layer to send IP options	RFC 1122: 3.2.1.8	M	5.3.1.3.3	m	Yes
op2b	Pass all IP options received to higher layer	RFC 1122: 3.2.1.8	M		m	Yes
op3b	IP layer silently ignore unknown options	RFC 1122: 3.2.1.8	M		m	Yes
Source Route Option:						
sr1	Originate and terminate Source Route Options	RFC 1122: 3.2.1.8(c)	M	5.3.1.3.3	m	Yes
sr2	Datagram with completed SR passed up to transport layer	RFC 1122: 3.2.1.8(c)	M		m	Yes
sr3	*Build correct (non-redundant) return route*	RFC 1122: 3.2.1.8(c)	M		m	Yes
sr4	Send multiple SR options in one header	RFC 1122: 3.2.1.8(c)	X		x	No
Routing Outbound Datagrams:						
ro1	Use address mask in local/remote decision	RFC 1122: 3.3.1.1	M	5.3.1.3.2.1.1	m	Yes
ro2	Operate with no gateways on connection network	RFC 1122: 3.3.1.1	M		m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
ro3	Maintain "route cache" of next-hop gateways	RFC 1122 3.3.1.2	M	5.3.1.3.2.1.1	m	Yes
ro4	Treat Host and Net Redirect the same		O		m	Yes
ro5	If no cache entry, use default gateway		M		m	Yes
ro51	Support multiple default gateways		M		m	Yes
ro6	Provide table of static routes		O		o	Yes No
ro61	Flag: route overridable by Redirects		O		o	Yes No
ro7	Key route cache on host, not net address	RFC 1122 3.3.1.3	O		o	Yes No
ro8	Include TOS in route cache	RFC 1122 3.3.1.3	O		m	Yes
ro9	Able to detect failure in next-hop gateway	RFC 1122 3.3.1.4	M		m	Yes
ro10	Assume route is good forever	RFC 1122 3.3.1.4	O		x	No
ro11	Ping gateways continuously	RFC 1122 3.3.1.4	X		x	No
ro12	Ping only when traffic is being sent	RFC 1122 3.3.1.4	M		m	Yes
ro13	Ping only when no positive indication	RFC 1122 3.3.1.4	M		m	Yes
ro14	Higher and lower layers give advice	RFC 1122 3.3.1.4	O		m	Yes
ro15	Switch from failed default gateway to another	RFC 1122 3.3.1.5	M		m	Yes
ro15a	If the current gateway fails, the host shall use round-robin selection of a new default gateway from its list of alternative gateways	RFC 1122 3.3.1.5	M		m	Yes
ro16	Manual method of entering config info	RFC 1122 3.3.1.6	M		m	Yes
Reassembly and fragmentation						
rf1	Able to reassemble incoming datagrams	RFC 1122 3.3.2	M	5.3.1.3.2.2	m	Yes
rf11	At least 576 byte datagrams		M		m	Yes
rf12	EMTU_R configurable or indefinite		O		m	Yes
rf121	EMTU_R equal or greater than MTU of connected network(s)		O		m	Yes
rf2	Transport layer able to learn MMS_R		M		m	Yes
rf3	Send ICMP Time Exceeded Message on reassembly timeout	RFC 1122 3.3.2	M		m	Yes
rf31	Fixed reassembly timeout value	RFC 1122 3.3.2	O		m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	*Status*	*Clause*	*Status*	
rf4	Pass MMS_S to higher layers	RFC 1122: 3.3.3	M	5.3.1.3.2.2	m	Yes
rf5	Local fragmentation of outgoing packets	RFC 1122: 3.3.3	O		o	Yes No
rf51	Else do not send bigger than MMS_S	RFC 1122: 3.3.3	M		m	Yes
rf6	Send max 576 to off-net destination	RFC 1122: 3.3.3	O		m	Yes
rf7	All-Subnets-MTU configuration flag	RFC 1122: 3.3.3	O		o	Yes No
Multihoming:						
mh1	Reply with same address as specific destination address	RFC 1122: 3.3.4.2	O	5.3.1.3.2.1.2	m	Yes
mh2	Allow application to choose local IP address	RFC 1122: 3.3.4.2	M		m	Yes
mh3	Silently discard datagram in "wrong" interface	RFC 1122: 3.3.4.2	O		o	Yes No
mh4	Only send datagram through "right" interface	RFC 1122: 3.3.4.2	O		o(1)	Yes No
Source Route Forwarding:						
sf1	Forward datagram with Source Route option	RFC 1122: 3.3.5	O	5.3.1.3.2.1.3	o	Yes No
sf11	Obey corresponding gateway rules	RFC 1122: 3.3.5	sf1:M		sf1:m	Yes N/A
sf111	Update TTL by gateway rules	RFC 1122: 3.3.5	sf1:M		sf1:m	Yes N/A
sf112	Able to generate ICMP error code 4,5	RFC 1122: 3.3.5	sf1:M		sf1:m	Yes N/A
sf113	*IP source address not local host*	RFC 1122: 3.3.5	sf1:O		sf1:o	Yes No
sf114	Update Timestamp, Record Route options	RFC 1122: 3.3.5	sf1:M		sf1:m	Yes N/A
sf12	Configurable for non-local SRing	RFC 1122: 3.3.5	sf1:M		sf1:m	Yes N/A
sf121	Defaults to OFF	RFC 1122: 3.3.5	sf1:M		sf1:m	Yes N/A
sf13	Satisfy gateway access rules for non-local SRing	RFC 1122: 3.3.5	sf1:M		sf1:m	Yes N/A

Note 1 Unless has embedded gateway functionality or is source routed

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
sf14	If not forward, send Destination Unreachable (cd 5)	RFC 1122 3.3.5	sf1:O		sf1:m[2]	Yes N/A
Broadcast:						
br1	Broadcast address as IP source address	RFC 1122 3.2.1.3	X	5.3.1.3.2.1.4	x	No
br2	Receive 0 or -1 broadcast formats OK	RFC 1122 3.3.6	O		m	Yes
br3	Configurable option to send 0 or -1 broadcast	RFC 1122 3.3.6	O		o	Yes No
br31	Default to -1 broadcast	RFC 1122 3.3.6	O		m	Yes
br4	Recognize all broadcast address formats	RFC 1122 3.3.6	M		m	Yes
br5	Use IP broadcast/multicast address in link layer broadcast	RFC 1122 3.3.6	M		m	Yes
br6	Silently discard link layer only broadcast datagrams	RFC 1122 3.3.6	O		m	Yes
br7	Use limited broadcast address for connected network	RFC 1122 3.3.6	O		m	Yes
Multicast/multi-addressing:						
mc1	Support local IP multicast/multi-addressing (RFC 1112)	RFC 1122 3.3.7	O	5.3.1.3.2.1.5	o	Yes No
mc2	Support IGMP (RFC 1112)	RFC 1122 3.3.7	O		o	Yes No
mc21	Support IGMP Messages	RFC 1122 3.3.7	mc2:O	5.4.1.1	mc2:m	Yes N/A
ver	version	RFC 1112 Appendix I	O	5.4.1.3	mc2:m	Yes N/A
typ	type	RFC 1112 Appendix I	O	5.4.1.4	mc2:m	Yes N/A
qury	Host Membership Query	RFC 1112 Appendix I	O		mc2:m	Yes N/A
rprt	Host Membership Report	RFC 1112 Appendix I	O		mc2:m	Yes N/A
unsd	unused (set to "0" on send, ignore upon recalpt)	RFC 1112 Appendix I	O	5.4.1.5	mc2:m	Yes N/A
chks	checksum	RFC 1112 Appendix I	O	5.4.1.6	mc2:m	Yes N/A

Note 2 This requirement is overruled if datagram is an ICMP error message

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
grpad	group address (class "D")	RFC 1122: 3.3.7	O	5.4.1 7	mc2:m	Yes N/A
mc3	Join all hosts group at startup	RFC 1122: 3.3.7	O	5.3.1.3 2.1.5	mc2:m	Yes N/A
mc4	Higher layers learn interface multicast capability	RFC 1122: 3.3.7	O		mc2:m	Yes N/A

A.4.4 PICS Proforma for Internet Control Message Protocol (ICMP)

A.4.4.1 Identification

Supplier	
Contact Point for Queries	
Implementation name(s) and Versions	
Other Information Necessary for full identification - eg, name(s) and version(s) for machines and/or operating systems; System Name(s)	

A.4.4.2 Protocol Summary

Protocol Version	
Addenda Implemented	
Amendments Implemented	
Have any exceptions been required? (Note: A YES answer means that the implementation does not conform to the *Internet Control Message Protocol*. Non-supported mandatory capabilities are to be identified in the PICS, with an explanation of why the implementation is non-conforming.	Yes_____ No_____
Date of Statement	

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A.4.4.3 Interfaces

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
iui	ICMP Uses IP	RFC 792	M	5.3.2.2	m	Yes

A.4.4.4 PDU Structure

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
mf	Message Format	RFC 792	M	5.3.2.2	m	Yes
vs	version 4		M		m	Yes
ihl	Internet Header Length		M		m	Yes
htos	Type of Service = 0		M		m	Yes
totl	Total Length in octets		M		m	Yes
id	Identification field		M		m	Yes
flg	Flags		M		m	Yes
frgo	Fragmentation Offset		M		m	Yes
httl	Time to Live		M		m	Yes
prot	Protocol - ICMP = 1		M		m	Yes
hchk	Checksum		M		m	Yes
sadd	Source Address		M		m	Yes
dadd	Destination Address		M		m	Yes

A.4.4.5 ICMP Message Formats

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
du	Destination Unreachable	RFC 792 RFC 1122 3.2.2.1	M	5.3.2.3.1	m	Yes
duda	Destination Address	RFC 792	M		m	Yes
dut3	Type = 3		M		m	Yes
duc	Code (0 or 1)		M		m	Yes
ducks	Checksum		M		m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
duih	Internet Header		M	5.3.2 3.1	m	Yes
du64	64 bits of data datagram		M		m	Yes
tem	Time Exceeded Message Gateway: Host Send: Host receive:	RFC 792 RFC 1122:3.2.2.4	M X M		m x m	Yes No Yes
temda	Destination address		M		m	Yes
temtt	Type = 11	RFC 792	M		m	Yes
temcd	Code (0 or 1)	RFC 792	M		m	Yes
temchk	Checksum	RFC 792	M		m	Yes
temih	Internet Header	RFC 792	M		m	Yes
tem64	64 bits of data datagram	RFC 792	M		m	Yes
ppm	Parameter Problem Message	RFC 792 RFC 1122: 3.2.2.5	M		m	Yes
ppmda	Destination address	RFC 792	M		m	Yes
ppmtt	Type = 12	RFC 792	M		m	Yes
ppmcc	Code (0 or 1)	RFC 792	M		m	Yes
ppmck	Checksum	RFC 792	M		m	Yes
ppmp	Pointer	RFC 792	M		m	Yes
ppmih	Internet Header	RFC 792	M		m	Yes
ppm64	64 bits of data datagram	RFC 792	M		m	Yes
sqm	Source Quench Message	RFC 792 RFC 1122: 3.2.2.3	M		m	Yes
sqmda	Destination address	RFC 792	M		m	Yes
sqmtt	Type = 4	RFC 792	M		m	Yes
sqmcc	Code = 0	RFC 792	M		m	Yes
sqmck	Checksum	RFC 792	M		m	Yes
sqmih	Internet Header	RFC 792	M		m	Yes
sqm64	64 bits of data datagram	RFC 792	M		m	Yes
rm	Redirect Message Gateway: Host send: Host receive:	RFC 792 RFC 1122: 3.2.2.2	M X M		m x m	Yes No Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
rmda	Destination address	RFC 792 RFC 1122 3.2.2.2	M	5.3.2.3.1	m	Yes
rmtt	Type = 5	RFC 792	M	5.3.2.3.1	m	Yes
rmcc	Code (0 to 3)	RFC 792	M		m	Yes
rmcks	Checksum	RFC 792	M		m	Yes
rmgia	Gateway Internet Address	RFC 792	M		m	Yes
rmih	Internet Header	RFC 792	M		m	Yes
rm64	64 bits of data datagram	RFC 792	M		m	Yes
emm	Echo Message	RFC 792 RFC 1122 3.2.2.6	M	5.3.2.3.2	m	Yes
emmda	Destination address	RFC 792 RFC 1122 3.2.2.6	M	5.3.2.3.1	m	Yes
emmtt	Type = 8	RFC 792	M		m	Yes
emmcc	Code = 0	RFC 792	M		m	Yes
emcks	Checksum	RFC 792	M		m	Yes
emmid	Identifier	RFC 792	M		m	Yes
emmsn	Sequence number	RFC 792	M		m	Yes
erm	Echo Reply Message	RFC 792 RFC 1122 3.2.2.6	M		m	Yes
ermsa	Source Address	RFC 792 RFC 1122 3.2.2.6	M		m	Yes
ermtt	Type = 0	RFC 792	M		m	Yes
ermcc	Code = 0	RFC 792	M		m	Yes
ermck	Checksum	RFC 792	M		m	Yes
ermid	Identifier	RFC 792	M		m	Yes
ermsn	Sequence number	RFC 792	M		m	Yes
tsp	Timestamp Message	RFC 792 RFC 1122 3.2.2.8	M		m	Yes
tspda	Destination address	RFC 792 RFC 1122 3.2.2.8	M		m	Yes
tsptt	Type = 13	RFC 792	M		m	Yes
tspcc	Code = 0	RFC 792	M		m	Yes
tspck	Checksum	RFC 792	M		m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
tspid	Identifier	RFC 792	M		m	Yes
tspsn	Sequence number	RFC 792	M		m	Yes
tsr	Timestamp Reply Message	RFC 792 RFC 1122: 3.2.2.8	M	5.3.2.3.2	m	Yes
tsrsa	Source Address	RFC 792 RFC 1122: 3.2.2.8	M		m	Yes
tsrtt	Type = 14	RFC 792	M		m	Yes
tsrcc	Code = 0	RFC 792	M		m	Yes
tsrck	Checksum	RFC 792	M		m	Yes
tsrid	Identifier	RFC 792	M		m	Yes
tsrsn	Sequence number	RFC 792	M		m	Yes
adrms	Address Mask Request/Reply	RFC 950: Appendix I RFC 1122: 3.2.2.9	M		m	Yes
irm	Information Request Message	RFC 792 RFC 1122: 3.2.2.7	X		x	No
irrm	Information Reply Message	RFC 792 RFC 1122: 3.2.2.7	X		x	No

A.4.4.6 ICMP Procedure

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
sd	Silently discard ICMP msg with unknown type	RFC 1122: 3.2.2	M	5.3.2.3	m	Yes
oc	Include more than 8 octets of orig datagram		O		o	Yes No
oc1	Included octets same as received		M		m	Yes
dm	Demux ICMP error to transport protocol		M		m	Yes
stos	Send ICMP error message with TOS=0		O		m	Yes
si	Send ICMP error message for:					
si1	ICMP error message	RFC 1122:3.2.2	X	5.3.2.3.1.1	x	No
si2	IP broadcast or IP multicast/multiaddress		X		x	No

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
si3	Link layer broadcast	RFC 1122 3.2.2	X		x	No
si4	Non-initial fragment		X		x	No
si5	Datagram with non-unique source address		X	5.3.2.3.1.1	x	No
ncm	Return ICMP error messages	RFC 1122 3.3.8	M		m	Yes
Destination Unreachable:						
du1	Generate Dest. Unreachable (code 2/3)	RFC 1122 3.2.2.1	O	5.3.2.3.1	m	Yes
du2	Pass ICMP Dest. Unreachable to higher layer	RFC 1122 3.2.2.1	M		m	Yes
du3	Higher layer act on Dest. Unreachable	RFC 1122 3.2.2.1	O		m	Yes
du31	Interpret Destination Unreachable as only hint	RFC 1122 3.2.2.1	M		m	Yes
Redirect:						
rd1	Host send redirect	RFC 1122 3.2.2.2	X	5.3.2.3.1	x	No
rd2	Update route cache when receive Redirect	RFC 1122 3.2.2.2	M		m	Yes
rd3	Handle both host and Net redirect	RFC 1122 3.2.2.2	M		m	Yes
rd4	Discard illegal redrect	RFC 1122 3.2.2.2	O	5.3.2.3.1	m	Yes
Source Quench:						
sq1	Send Source Quench if buffering exceeded	RFC 1122 3.2.2.3	O	5.3.2.3.1	o	Yes No
sq2	Pass Source Quench to higher layer	RFC 1122 3.2.2.3	M		m	Yes
sq3	Higher layer act on Source Quench	RFC 1122 3.2.2.3	O		m	Yes
Time Exceeded:						
tmxe	Time Exceeded: pass to higher layer	RFC 1122 3.2.2.4	M		m	Yes
Parameter Problem:						
pp1	Host send parameter problem messages	RFC 1122 3.2.2.5	O	5.3.2.3.1	m	Yes
pp2	Pass Parameter Problem to higher layer	RFC 1122 3.2.2.5	M		m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
pp3	Report parameter problem to user	RFC 1122: 3.2.2.5	O	5.3.2.3.1	o	Yes No
Echo request or Reply:						
ec1	Echo server and echo client	RFC 1122: 3.2.2.6	M	5.3.2.3.2	m	Yes
ec2	Echo client	RFC 1122: 3.2.2.6 .	O		m	Yes
ec3	Discard Echo Request to broadcast address	RFC 1122: 3.2.2.6	O		o	Yes No
ec4	Discard Echo Request to multicast address	RFC 1122: 3.2.2.6	O		o	Yes No
ec5	Use specific-dest address as Echo-Reply source	RFC 1122: 3.2.2.6	M		m	Yes
ec6	Send same data in Echo Reply	RFC 1122: 3.2.2.6	M		m	Yes
ec7	Pass Echo Reply to higher layer	RFC 1122: 3.2.2.6	M		m	Yes
ec71	If fragmentation of Echo Reply is required but not implemented, truncate datagram to max transmission size	RFC 1122: 3.2.2.6	M		m	Yes
ec8	Reflect Record Route, Time Stamp options	RFC 1122: 3.2.2.6	O		m	Yes
ec9	Reverse and reflect Source Route option	RFC 1122: 3.2.2.6	M		m	Yes
Information Request/Reply:						
irr	ICMP Information Request or Reply:	RFC 1122: 3.2.2.7	O	5.3.2.3.2	x	No
Timestamp and Timestamp Reply:						
itr	Host implement Timestamp and Timestamp Reply:	RFC 1122: 3.2.2.8	O	5.3.2.3.2	o	Yes No
itr1	Minimize delay variability	RFC 1122: 3.2.2.8	itr:O		itr:m	Yes N/A
itr2	Silently discard broadcast Timestamp	RFC 1122: 3.2.2.8	itr:O		itr:o	Yes No
itr3	Silently discard multicast Timestamp	RFC 1122: 3.2.2.8	itr:O		itr:o	Yes No
itr4	Use specific-dest address as TS reply source	RFC 1122: 3.2.2.8	itr:M		itr:m	Yes N/A
itr5	Reflect Record Route, Time Stamp Options	RFC 1122: 3.2.2.8	itr:O		itr:m	Yes N/A
itr6	Reverse and reflect source route option	RFC 1122: 3.2.2.8	itr:M		itr:m	Yes N/A

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
itr7	Pass timestamp reply to higher layer	RFC 1122 3.2.2.8	itr:M		itr m	Yes N/A
itr8	Obey rules for "standard value"	RFC 1122 3.2.2.8	itr:M		itr m	Yes N/A
Address Mask Request and Reply:						
iar1	Address Mask Method is Configurable	RFC 1122 3.2.2.9	M	5.3.2.3.2	m	Yes
iar2	Support static configuration of address mask		M		m	Yes
iar3	Get address mask dynamically during booting		O		o	Yes No
iar4	Host Send ICMP Address Mask Request/receive Address Mask Reply		O		o	Yes No
iar41	Retransmit Address Mask Req. if no reply		iar4 M		iar4 m	Yes N/A
iar42	Assume default mask if no reply		iar4:O		iar4:m	Yes N/A
iar43	Update address mask from first reply only		iar4 M		iar4:m	Yes N/A
iar5	Reasonableness check on Address Mask		O		m	Yes
iar6	Send unauthorized Address Mask Reply Message		X		x	No
iar61	Explicitly configured to be agent		M		m	Yes
iar7	Static Configuration ==> Addr-Mask-Authoritative flag		O		m	Yes
iar71	Broadcast Address Mask Reply when initiated		M		m	Yes

A.4.5 PICS Proforma for Combat Net Radio (CNR)

This paragraph, in conjunction with MIL-STD 2045-14502-6A, is mandatory for systems operating with CNR in the data channel. The physical, data link, and Intranet layers are described in MIL-STD 2045-14502-6A. These features, while mandatory for CNR, are not limited to CNR. They may be used in other applications if appropriate.

This paragraph includes the enhancements peculiar to CNR for the Internet Protocol, the Transmission Control Protocol (TCP), and the User Datagram Protocol (UDP) contained in previous sections of Annex A. It includes use of message segmentation/reassembly in the layer above UDP.

Implementations of TCP, UDP and IP with CNR are intended to be compliant with extensions added at the UDP/Upper Layer Protocol (ULP) software interface to support direct specification of multiple addresses using the IP Option field. The interface between the Transport Layer UDP protocol and the ULP includes procedures for selective directed broadcast.

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A.4.5.1 Identification

Supplier	
Contact Point for Queries	
Implementation name(s) and Versions	
Other Information Necessary for full identification - eg, name(s) and version(s) for machines and/or operating systems; System Name(s)	

A.4.5.2 Protocol Summary

Protocol Version	
Addenda Implemented	
Amendments Implemented	
Have any exceptions been required? (Note: A YES answer means that the implementation does not conform to the CNR modifications to the Transmission Control Protocol/ Internet Protocol. Non-supported mandatory capabilities are to be identified in the PICS, with an explanation of why the implementation is non-conforming.	Yes_____ No_____
Date of Statement	

A.4.5.3 Message Segmentation Capabilities

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
msrp	Message Segmentation/Reassembly Protocol			5.5.1.2	m	Yes
mss	Maximum segment size defined			5.5.1.3.1	m	Yes
ack	End-to-end acknowledgment			5.5.1.3.2	m	Yes
hdr	Segmentation/reassembly header capability			5.5.1.3.3	m	Yes

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A.4.5.4 Segmentation/Reassembly Header

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
srcp	Source Port number	*	*	5.5.1.4	m	Yes
dstp	Destination Port number			5.5.1.5	m	Yes
typ	Type of Header			5.5.2	m	Yes
hlen	Header Length			5.5.2.1	m	Yes
mgsn	Message Serial Number			5.5.2.2	m	Yes
sgtn	Segment Number			5.5.2.3	m	Yes
lstsn	Last Segment Number			5.5.2.4	m	Yes
acbtm	Acknowledged segments bitmap			5.5.2.5	m	Yes
intly	Interface with other Layers			5.5.2.6	m	Yes
sdprc	Source device procedures			5.5.2.7	m	Yes
ddprc	Destination device procedures			5.5.2.8	m	Yes

* This PICS serves as the base standard for CNR

A.4.5.5 IP Option - Directed Broadcast Procedure

Procedures for source hosts, transmit routers and destination routers are provided below.

Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
ip22	User IP option field capability selected	RFC 1122, 3.2.1.8	M	5.5.3, 5.5.3.4	mulop:m	Yes N/A
ip22op	IP option header fields defined		*	5.5.3.3	m	Yes
ip22a	"Type" field contains copy flag, class, number of option.	*	*	5.5.3.3a	m	Yes
ip22b	Copy flag =1			5.5.3.3b	m	Yes
ip22c	Class field set to zero			5.5.3.3c	m	Yes
ip22d	LENGTH set to 2 + 4*(IP addresses)			5.5.3.3d	m	Yes
ip22e	Each IP option address occupies 4 octets			5.5.3.3e	m	Yes
dirbcst	Directed broadcast address formed from Net ID for that network. The DB address is used as the Destination Address in the IP packet			5.5.3.5.a.2	m	Yes

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Item	Protocol Feature	Base Standard		Profile		Support
		Reference	Status	Clause	Status	
netids	Group destination host internet addresses identified by Net IDs. For N distinct Net IDs, there will be at least N distinct directed broadcast (DB) packets.	*	*	5.5.3.5.a.1	m	Yes
optfld	Place the entire IP address in the group in the Multi-address IP option field. The total length of all IP options is limited. If other options are included in the IP header, the number of addresses in the multi-address option may need to be reduced.			5.5.3.5.a.3	m	Yes
optfld1	Header not to exceed 38 octets			5.5.3.3	m	Yes
trsm	Transmit the completed packets normally, i.e. network compatibility must be noted and assured			5.5.3.5.a.4	m	Yes
Transit router procedure						
trsrtr	Transit routers route the packets normally	*	*	5.5.3.5.b	m	Yes
	Destination gateway procedure:					
itahdr	Form an Intranet protocol header containing the destination host data link addresses and the relay adresses			5.5.3.5.b.1	m	Yes
trst	Transmit packet within network/subnetwork			5.5.3.5.b.2	m	Yes

* This PICS serves as the base standard for CNR

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ANNEX B (informative)

CONCLUDING MATERIAL

B.1 Deviations from the Base Standards/Referenced Profiles

This MIL-STD documents the data contained in the IAB standards 3, 5, 6, and 7 in the ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" in MIL-HDBK-829A format. This DSP does not deviate from the protocol as written in the RFC base standards, except where indicated in this DSP.

The classification of the requirements in RFCs have been changed in the DSPICS to the following:

RFC	(MIL-STD) DSPICS
MUST	Mandatory
SHOULD	Mandatory
MAY	Optional
SHOULD NOT	Prohibited
MUST NOT	Prohibited

B.2 Subject Term (Keyword) Listing

IAB 5
IAB 7
IAB 6
IAB 3
ICMP
Internet Protocol
Internet Activities Board
Internet Message Control Protocol
IP
Network
RFC
TCP
Transmission Control Protocol
Transport
UDP
User Datagram Protocol

B.3 Preparing Activity:

Defense Information Systems Agency (DISA) - DC
Project: DCPS-0013, SP 02

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B.4 Reviewing Activity:

Army	SC, PT
Air Force	3, 17, 29, 33, 90
DLA	DH
DMA	MP
DIA	DI
DOT	OST
NSA	NS
OASD	IQ, DO, MA, IR
ODISC4	AC
NAVY	EC, CH, ND, TD, OM
USMC	MC, CG

B.5 Custodians:

DISA:	DC
Army:	SC
Air Force:	90
Navy:	OM
DIA:	DI
NSA:	NS
USMC:	MC
DLA:	DH
Other:	Joint Staff/Architecture & Integration, USSPACECOM

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1,2, 3, and 8. In block 1, both the document number and revision letter should be given.
2 The submitter of this form must complete blocks 4, 5, 6, and 7.
3 The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-STD 2045-14502-1A	2 DOCUMENT DATE (YYMMDD) 950727

3 DOCUMENT TITLE Information Technology - DOD Standardized Profile - Internet Transport Profile for DOD Communications - Part 1: Transport and Internet Services

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) DSN *(If applicable)*	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY DEFENSE INFORMATION SYSTEMS AGENCY (DISA)

a NAME Frank Curcio	b TELEPHONE (Include Area Code) (1) Commercial 908-532-7732 (2) DSN 992-
c ADDRESS (Include Zip Code) Director JIEO Attn: JEBBF Ft. Monmouth, NJ 07703-5613	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041- 3466 Telephone (703) 756-2340 DSN 289-2340

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